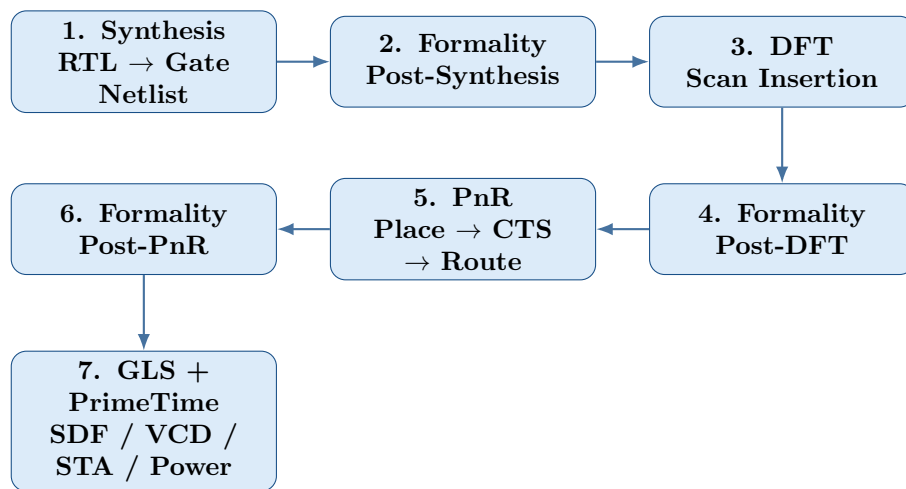


Backend Implementation Flow

UART-Controlled Low-Power Multi-Clock Digital System

Purpose. This report summarizes **what was done in the backend flow**, not the folder structure. The implementation was taken from RTL through Synthesis, DFT, PnR, formal equivalence checks, Gate-Level Simulation, and PrimeTime signoff.

1. Complete Flow



Final status: Synthesis, DFT, all three Formality checkpoints, PnR timing closure, physical verification, post-PnR GLS, and PrimeTime setup/hold checks reached passing results.

2. Synthesis Results

Metric	Result
Total cell area	22,193.74 μm^2
Number of cells	1,944
Combinational / Sequential	1,537 / 369 cells
Worst reported setup slack	+4.10 ns - MET
Worst reported hold slack	+0.65 ns - MET

What happened here: The RTL was synthesized into a technology-mapped gate-level netlist using the TSMC 130 nm standard-cell library. Area, clocks, constraints, setup, hold, and power reports were generated.

3. Post-Synthesis Formality

Verification SUCCEEDED - 363 passing compare points, with **0 failing** compare points. This confirmed that synthesis preserved the functional behavior of the RTL.

4. DFT Results

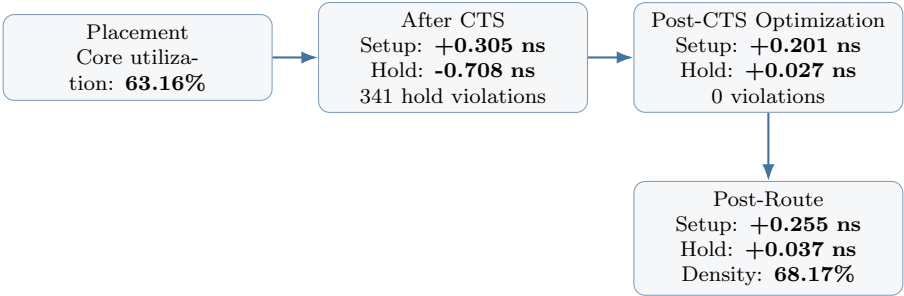
Metric	Result
DFT test coverage	99.69%
Total cell area after DFT	22,196.09 μm^2
Worst reported DFT setup slack	+77.07 ns - MET
Worst reported DFT hold slack	+0.31 ns - MET

What happened here: Scan/DFT structures and test controls were inserted, then DFT coverage and timing were checked before moving to physical implementation.

5. Post-DFT Formality

Verification SUCCEEDED - final Formality log reports **360 passing compare points** and **0 failing** points, confirming equivalence under the applied DFT verification setup.

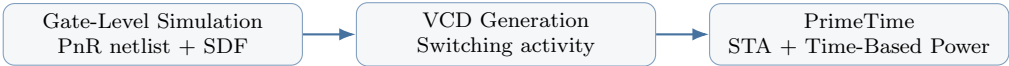
6. Physical Design - PnR



Timing closure: CTS introduced hold violations, then post-CTS optimization removed all **341** hold violations while keeping setup positive. After routing, both setup and hold remained clean with **0 violating paths**.

Final PnR Check	Result
Max capacitance / transition / fanout	0 / 0 / 0 violations
Geometry / DRC	0 violations
Connectivity	0 violations, 0 warnings
Process antenna	0 violations
Post-PnR Formality	SUCCEEDED - 360 passing, 0 failing

7. Post-PnR GLS and PrimeTime Signoff



Signoff Item	Result
SDF back-annotation	Successfully completed
Gate-Level Simulation	TEST PASSED
VCD annotation in PrimeTime	100% nets and 100% leaf cells annotated
PrimeTime setup violations	0
PrimeTime hold violations	0
Worst PrimeTime setup slack	+0.25 ns
Worst PrimeTime hold slack	+0.23 ns
Time-based total power	3.478e-04 (PrimeTime report value)